

Section: Particles and Waves

Topic: Wave-Particle Duality

Light is incident on a clean metal surface in a vacuum.
Electrons are released from the surface by the photoelectric effect.

(a) Name the type of particle that is the source of the photons in the light.

✔ The source of photons in light is the **electron**.

Electrons in atoms **drop to lower energy levels** and emit photons in the process.

(b) The maximum kinetic energy of the emitted electrons is 1.8×10^{-19} J.

The work function of the metal is 3.4×10^{-19} J.
Calculate the frequency of the incident radiation.

Use the photoelectric equation:

$$E_k = hf - \phi \quad \Rightarrow \quad hf = E_k + \phi$$

$$hf = 1.8 \times 10^{-19} + 3.4 \times 10^{-19} = 5.2 \times 10^{-19} \text{ J}$$

Now solve for f :

$$f = \frac{E}{h} = \frac{5.2 \times 10^{-19}}{6.63 \times 10^{-34}} = 7.84 \times 10^{14} \text{ Hz}$$

(c) Describe how the photoelectric effect supports the idea that light can behave as a particle.

✔ In the photoelectric effect:

- **Electrons are ejected instantly** when the light is above a certain frequency (no time delay).
- The energy of the electrons depends on the **frequency**, not the intensity of the light.
- These observations suggest that light arrives in **packets (photons)**, each with energy $E = hf$.

This supports the concept that light behaves as **particles** (photons), not just as waves.

🧠 Revision Tips

- Photoelectric equation:

$$E_k = hf - \phi$$

- ϕ is the **work function**: the energy needed to free an electron
- h is Planck's constant: 6.63×10^{-34} Js
- The photoelectric effect demonstrates **quantisation** of energy — a hallmark of particle behaviour